RESEARCH LETTER

# Channel attention gives no transferable gain in few-label hyperspectral CNNs

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**ABSTRACT**
Channel-attention blocks are often added to lightweight networks for hyperspectral image classification, usually with gains reported on random pixel splits. We test whether any such gain transfers. Seven modulation blocks (SE, ECA, CBAM, GCT, SRM and two variants of a dispersion-aware block we introduce, SNSM) are compared on one backbone with one training protocol, no test-based model selection, up to fifteen paired seeds, bootstrap intervals and Holm-corrected Wilcoxon tests. On random 5% splits, blocks differ only on Indian Pines, where the two dispersion-based gates (SNSM, SRM) exceed the others by 1.8–3.8 points of average accuracy (AA) while overall accuracy differs by under 0.2 points. The gap comes from three classes with one or two training pixels: giving only those classes 15 pixels shrinks it to 0.6 points. Imposing the same scarcity on Pavia University and Kennedy Space Center (KSC) does not recreate it, and on patch-disjoint splits accuracy falls by 32–34 points with mostly non-significant differences. We also show that the distributed KSC file stores small negative values as unsigned integers; reading them as signed raises seven convolutional variants from 77–92% to over 99% AA.

**KEYWORDS**
Hyperspectral image classification; channel attention; spatial leakage; evaluation protocol; few labels

## 1. Introduction

Hyperspectral images record hundreds of contiguous bands. This makes pixel-level land-cover classification possible, but when labelled samples are few it exposes classifiers to the Hughes phenomenon (Hughes 1968; Bioucas-Dias et al. 2013). Convolutional neural networks (CNNs) that operate on spatial–spectral patches are the usual lightweight choice (Li et al. 2019; Audebert, Le Saux, and Lefevre 2019; Makantasis et al. 2015; Chen et al. 2016; Zhong et al. 2018; Roy et al. 2020), and transformers add global context at a higher cost (Hong et al. 2022; Sun et al. 2022). Channel attention is a cheap way to make such networks input-dependent: squeeze-and-excitation (SE) predicts one scale per channel from pooled means through fully connected (FC) layers (Hu, Shen, and Sun 2018); ECA replaces the FC layers by a one-dimensional convolution (Wang et al. 2020); CBAM adds a spatial gate (Woo et al. 2018); GCT normalises channel statistics and gates them with $1 + \tanh(\cdot)$ (Yang et al. 2020); and

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SRM pools channel means and standard deviations (Lee, Kim, and Nam 2019). Hyperspectral networks have adopted these blocks widely (Mei et al. 2019; Haut et al. 2019; Li et al. 2020; Zhu et al. 2021; Roy et al. 2021).

Most reported gains come from random pixel splits. With patch-based models, training and test patches then overlap, so test accuracy partly measures memorised spatial context (Liang et al. 2017; Nalepa, Myller, and Kawulok 2019). This leakage has been surveyed (Feng et al. 2023), disjoint sampling changes the accuracy of deep models (Ahmad et al. 2024), and spatially separated splits lower the overall accuracy of CNNs and transformers on Pavia University by about 40 points (Alkhatib 2026). These studies compare whole architectures. They do not show whether the *choice of channel-attention block* matters, or under which conditions.

Our contributions are:

- a controlled comparison of seven modulation blocks on one backbone, with identical inputs and schedule, no test-based model selection, paired seeds, bootstrap intervals and Holm-corrected Wilcoxon tests, under random 5% splits, splits that change the training counts of all or only the smallest classes, the standard Indian Pines split, and patch-disjoint splits in which no test patch overlaps a training patch (Sections 2–3);
- the finding that no block gives a gain that transfers: the one effect we find, dispersion-based gates on Indian Pines, is caused by three classes with one or two training pixels, and it does not reappear when the same scarcity is imposed on two other scenes (Section 4);
- SNSM, a dispersion-aware block used here as a probe; its ablation shows that the channel variance, which SRM also uses, carries the Indian Pines gain;
- a diagnosis of the Kennedy Space Center (KSC) data encoding, which earlier work treated as sensor saturation (Kakde et al. 2018), and its effect on CNN accuracy.

## 2. Blocks compared

All blocks are inserted in the same backbone. A $3 \times 3$ convolution maps the $B$ input features to $C = 64$ channels (batch normalisation (Ioffe and Szegedy 2015), ReLU). Two residual blocks (He et al. 2016) follow, each computing

$$\mathbf{h} = \mathrm{ReLU}(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathbf{x}))), \qquad \mathbf{y} = \mathrm{ReLU}\big(\mathrm{BN}(\mathrm{Conv}_{3\times3}(\mathcal{M}(\mathbf{h}))) + \mathbf{x}\big), \quad (1)$$

where $\mathcal{M}$ is the block under study ($\mathcal{M}$ = identity for the plain CNN). Global average pooling, dropout (0.4) (Srivastava et al. 2014) and a linear layer produce the class scores. SE, ECA (kernel 3, the adaptive value for 64 channels) and CBAM (reduction 16, $7 \times 7$ spatial gate) output a sigmoid scale per channel; GCT uses its $\ell_2$ embedding and an identity start; SRM gates each channel from its mean and standard deviation through a channel-wise FC layer and batch normalisation.

**SNSM.** For $\mathbf{h} \in \mathbb{R}^{C\times H\times W}$, SNSM computes for every channel $c$ the spatial mean $\mu_c$, the spatial variance $v_c$ and the cross-channel gradient $g_c = (\mu_{c+1} - \mu_{c-1})/2$. Each descriptor vector is standardised across channels, the three are stacked into $\mathbf{S} \in \mathbb{R}^{3\times C}$, and two one-dimensional convolutions along the channel axis (8 hidden channels, kernel

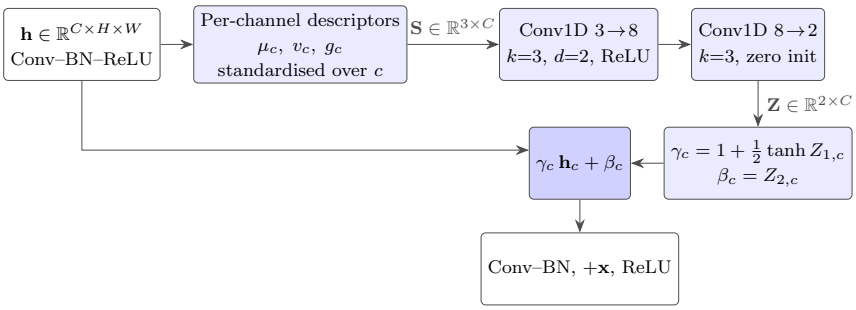



**Figure 1.** SNSM inside a residual block. Per-channel descriptors of the feature map $\mathbf{h}$ are filtered along the channel axis by two one-dimensional convolutions, whose output sets a bounded scale $\gamma_c$ and a shift $\beta_c$ for each channel. The second convolution starts at zero, so the block is initially the identity.

3, dilation 2 in the first, replicate padding) give $\mathbf{Z} \in \mathbb{R}^{2\times C}$. The modulation is

$$\gamma_c = 1 + \tfrac{1}{2}\tanh(Z_{1,c}), \qquad \beta_c = Z_{2,c}, \qquad \mathcal{M}(\mathbf{h})_c = \gamma_c\,\mathbf{h}_c + \beta_c. \tag{2}$$

The second convolution starts at zero, so the block is initially the identity; it adds 130 parameters per block (Figure 1). SNSM ($\gamma$) drops the shift. Like SRM it sees channel dispersion; unlike SRM it filters the descriptors jointly across neighbouring feature channels and can shift as well as scale. Including it with and without each descriptor lets us test which channel statistic matters.

## 3. Data and protocols

**Data.** We use Indian Pines (IP), Pavia University (PU), KSC and Botswana (BW) (Baumgardner, Biehl, and Landgrebe 2015; Gamba 2004; Grupo de Inteligencia Computacional 2020) (Table 1). Each band is smoothed with a $5\times5$ Wiener filter, principal component analysis (PCA) (Jolliffe and Cadima 2016), fitted without labels on all pixels, keeps 30 components, and the db4 wavelet approximation of each component (Mallat 1989) is resized and appended, giving $B = 60$ features. Each labelled pixel is represented by its $15 \times 15 \times B$ patch.

**KSC encoding.** The commonly distributed KSC file is stored as unsigned 16-bit integers, but it contains small negative values ($-27$ to $-1$) that are stored in two's complement and are therefore read as 65,509–65,535; about 5% of its pixels contain at least one such value. Kakde et al. (2018) noticed these values, described them as saturation and set them to zero. The DeepHyperX toolbox accompanying Audebert, Le Saux, and Lefevre (2019) loads the file unchanged and scales it by its global minimum and maximum, which compresses the valid range of 0–1,244 into less than 2% of the scale. We did not audit which published KSC results used such a loader. We read KSC as signed integers. Without this correction, the wrapped values dominate the PCA and seven CNN variants reach only 77–92% AA on KSC with random 5% splits, instead of 99.2–99.5%.

**Protocols.** *Random 5%*: for each class, 5% of the labelled pixels (at least one) train and the rest test, giving 505/9,744 (IP), 2,135/40,641 (PU), 256/4,955 (KSC) and 156/3,092 (BW) pixels; Alfalfa, Grass-pasture-mowed and Oats then have one or two training pixels. *Random 50/15* (IP): a random split with the class counts of the standard split, 15 training pixels for those three classes and 50 for each of the others (695/9,554). *Smallest 15* (IP): the random 5% split with only those three classes raised

**Table 1.** Benchmark scenes and the random 5% splits (seed 42; split sizes are the same for every seed).

| Scene | Sensor | Size (pixels) | Bands | GSD (m) | Classes | Labelled | Train/test (random) |
|---|---|---|---|---|---|---|---|
| Indian Pines (IP) | AVIRIS | 145 × 145 | 200 | 20 | 16 | 10,249 | 505/9,744 |
| Pavia University (PU) | ROSIS | 610 × 340 | 103 | 1.3 | 9 | 42,776 | 2,135/40,641 |
| Kennedy Space Center (KSC) | AVIRIS | 512 × 614 | 176 | 18 | 13 | 5,211 | 256/4,955 |
| Botswana (BW) | Hyperion | 1476 × 256 | 145 | 30 | 14 | 3,248 | 156/3,092 |




**Figure 2.** Patch-disjoint split of Pavia University (seed 42). Training pixels form one compact cluster per class; test pixels whose 15 × 15 patch would overlap a training patch are excluded.

to 15 training pixels. *Smallest 1/1/2* (PU, KSC): the random 5% split with the three smallest classes of each scene reduced to 1, 1 and 2 training pixels, as in IP. *Standard* (IP): the fixed training and test sets distributed with SpectralFormer (Hong et al. 2022) (695/9,671). Its training pixels come from separate regions, but with 15 × 15 patches every test patch still overlaps a training patch; the corresponding PU and Houston sets were no longer available from the authors' link. *Patch-disjoint* (PU, KSC): for each class, the 5% of its pixels nearest a random centre form a compact training cluster, and every test pixel within 14 pixels of any training pixel is discarded, so no test patch overlaps a training patch (Figure 2); 33,847–36,453 (PU) and 3,077–3,513 (KSC) test pixels remain, covering all 9 PU classes and at least 11 of the 13 KSC classes. The same rule leaves no test pixels for five IP classes, so IP is not evaluated this way.

**Training and statistics.** Every model sees the same features and splits for a given seed. Training uses cross-entropy, AdamW (Loshchilov and Hutter 2019) (learning rate $10^{-3}$, weight decay $10^{-4}$), step decay (factor 0.5 every 40 epochs), batch size 64 and exactly 100 epochs, without augmentation or early stopping; the final model is evaluated once, so no validation or test data are used for selection. All eight backbone variants use 15 seeds (42–56) under every protocol except the random 5% splits of PU, KSC and BW, which use five. On the standard split the seed changes only the

**Table 2.** Random 5% splits: OA and AA (%, mean ± std). The eight backbone variants use 15 seeds on IP and five elsewhere; FC ($\gamma$+$\beta$), HybridSN (Roy et al. 2020) and SSFTT (Sun et al. 2022) use five. Best mean per column in bold.

| Method | IP | | KSC | | PU | | BW | |
|---|---|---|---|---|---|---|---|---|
| | OA | AA | OA | AA | OA | AA | OA | AA |
| Plain CNN | 96.28±0.79 | 89.49±2.44 | 99.65±0.34 | 99.51±0.36 | **99.59±0.07** | 99.01±0.16 | 99.09±0.55 | 98.87±0.90 |
| SE ($\gamma$) | 96.34±0.62 | 90.04±2.18 | 99.61±0.28 | 99.43±0.38 | 99.53±0.19 | 98.80±0.56 | 98.95±0.82 | 98.76±1.07 |
| ECA ($\gamma$) | 96.28±0.68 | 90.87±2.53 | 99.61±0.47 | 99.40±0.66 | 99.58±0.10 | 99.00±0.20 | **99.13±0.76** | **98.90±1.08** |
| CBAM | 96.19±0.60 | 91.58±1.49 | 99.54±0.56 | 99.23±0.83 | 99.58±0.06 | **99.11±0.18** | 98.99±0.87 | 98.78±1.20 |
| GCT | 96.28±0.80 | 90.07±2.34 | **99.69±0.26** | **99.55±0.28** | 99.58±0.08 | 98.99±0.18 | 99.11±0.56 | 98.88±0.91 |
| SRM | 96.20±0.71 | 93.04±2.14 | 99.69±0.36 | 99.51±0.49 | 99.55±0.07 | 98.92±0.17 | 98.87±0.63 | 98.67±0.97 |
| SNSM ($\gamma$) | 96.17±0.73 | 92.89±2.00 | 99.69±0.26 | 99.50±0.39 | 99.49±0.05 | 98.91±0.16 | 99.10±0.62 | 98.89±0.96 |
| SNSM ($\gamma$+$\beta$) | 96.27±0.67 | **93.33±1.68** | 99.63±0.55 | 99.45±0.73 | 99.47±0.05 | 98.93±0.16 | 99.04±0.69 | 98.85±1.03 |
| FC ($\gamma$+$\beta$) | 96.24±0.52 | 92.51±1.60 | 99.63±0.40 | 99.48±0.47 | 99.51±0.06 | 98.88±0.14 | 98.80±0.70 | 98.63±0.98 |
| HybridSN | 88.30±4.91 | 73.64±8.28 | 95.00±3.02 | 91.20±5.32 | 97.84±1.52 | 95.46±4.12 | 77.16±37.92 | 75.56±38.31 |
| SSFTT | **96.71±0.79** | 92.35±1.81 | 99.38±0.28 | 99.15±0.32 | – | – | – | – |

**Table 3.** Indian Pines: AA (%, mean ± std over 15 seeds) of the eight backbone variants under three protocols. The last row is the range of the block means.

| Block | Random 5% | Random, 50/15 per class | Standard split |
|---|---|---|---|
| Plain CNN | 89.49±2.44 | 97.83±0.52 | 97.24±0.23 |
| SE ($\gamma$) | 90.04±2.18 | **97.90±0.44** | 97.25±0.26 |
| ECA ($\gamma$) | 90.87±2.53 | 97.79±0.49 | 97.15±0.22 |
| CBAM | 91.58±1.49 | 97.78±0.51 | 97.45±0.29 |
| GCT | 90.07±2.34 | 97.82±0.53 | 97.23±0.22 |
| SRM | 93.04±2.14 | 97.76±0.57 | **97.50±0.22** |
| SNSM ($\gamma$) | 92.89±2.00 | 97.77±0.54 | 97.37±0.22 |
| SNSM ($\gamma$+$\beta$) | **93.33±1.68** | 97.82±0.43 | 97.48±0.26 |
| Range of block means | 3.84 | 0.14 | 0.35 |

initialisation and batch order; elsewhere it also changes the split. For each scene and protocol we compare SNSM ($\gamma$+$\beta$) with the seven other blocks by paired same-seed differences, report a 95% percentile bootstrap interval of the mean difference, and apply two-sided Wilcoxon signed-rank tests with Holm correction over those seven comparisons. We call blocks practically indistinguishable when their mean AA differs by less than 0.5 points. Runs used PyTorch (Paszke et al. 2019) on NVIDIA Tesla T4 GPUs, except ten KSC patch-disjoint seeds, which ran on an RTX 3050 laptop GPU; each paired comparison uses runs from the same GPU.

## 4. Results

**Random 5% splits.** On PU, KSC and BW the eight backbone variants lie within 0.32 AA points of each other (Table 2). Only IP separates them: SNSM ($\gamma$+$\beta$) has the highest AA and is significantly better than the plain CNN, SE, ECA, CBAM and GCT, by 1.75–3.84 points (Table 4), but not than SRM or SNSM ($\gamma$). The overall accuracy of all eight variants differs by only 0.17 points, so the gain comes from a few classes. Oats and Grass-pasture-mowed have one training pixel and 19 and 27 test pixels, so a few test pixels move their per-class accuracy by tens of points. Indeed the gap lies almost entirely in these two single-pixel classes: SRM and the two SNSM variants reach 80.0–84.4% on Grass-pasture-mowed and 66.3–77.2% on Oats, against 62.7–78.5% and 36.8–56.1% for the other blocks, whereas on Alfalfa (two training pixels) all blocks lie within 3.3 points, and their mean accuracy over the remaining 13 classes differs by less than one point. HybridSN and SSFTT, trained under our protocol, are less accurate than the backbone variants.

**Class counts, not split geometry.** When every IP class has 15–50 training pixels,

**Table 4.** Paired AA differences SNSM ($\gamma$+$\beta$) − block (points) with 95% bootstrap intervals. *: Holm-corrected two-sided Wilcoxon $p < 0.05$ (Holm family: the seven comparisons in one column).

| SNSM ($\gamma$+$\beta$) − | IP random 5% | IP random 50/15 | IP standard | PU disjoint | KSC disjoint |
|---|---|---|---|---|---|
| Plain CNN | +3.84* [+2.6, +5.1] | -0.01 [-0.2, +0.2] | +0.24 [+0.1, +0.4] | +0.31 [-1.6, +2.2] | +3.46* [+1.7, +5.1] |
| SE ($\gamma$) | +3.29* [+2.4, +4.3] | -0.08 [-0.2, +0.0] | +0.23 [+0.1, +0.4] | -0.47 [-2.5, +1.4] | -0.19 [-2.1, +1.6] |
| ECA ($\gamma$) | +2.45* [+1.1, +3.9] | +0.04 [-0.1, +0.2] | +0.33* [+0.2, +0.5] | -0.39 [-2.2, +1.4] | +1.86 [+0.6, +3.3] |
| CBAM | +1.75* [+1.0, +2.5] | +0.05 [-0.1, +0.2] | +0.04 [-0.1, +0.2] | +1.03 [-0.6, +2.4] | +0.91 [-0.7, +2.4] |
| GCT | +3.26* [+2.2, +4.4] | +0.01 [-0.2, +0.2] | +0.25 [+0.1, +0.4] | +0.92 [-0.8, +2.5] | +3.51* [+1.8, +5.2] |
| SRM | +0.29 [-0.6, +1.3] | +0.07 [-0.1, +0.2] | -0.02 [-0.2, +0.1] | +1.33 [-0.3, +2.7] | +4.25* [+2.0, +6.6] |
| SNSM ($\gamma$) | +0.44 [-0.2, +1.1] | +0.06 [-0.1, +0.3] | +0.12 [-0.1, +0.3] | +0.03 [-1.7, +1.8] | +1.40 [-0.7, +3.6] |

**Table 5.** AA (%, mean ± std) of random 5% splits and of the same splits with only the three smallest classes changed (15 seeds; the unchanged PU and KSC splits use five). The last row is the range of the block means.

| Block | IP, 5% | IP, smallest 15 | PU, 5% | PU, smallest 1/1/2 | KSC, 5% | KSC, smallest 1/1/2 |
|---|---|---|---|---|---|---|
| Plain CNN | 89.49±2.44 | 96.03±0.76 | 99.01±0.16 | 84.20±3.43 | 99.51±0.36 | **93.12±3.44** |
| SE ($\gamma$) | 90.04±2.18 | 95.93±0.64 | 98.80±0.56 | 84.35±3.29 | 99.43±0.38 | 92.09±3.25 |
| ECA ($\gamma$) | 90.87±2.53 | 96.10±0.76 | 99.00±0.20 | 84.80±3.23 | 99.40±0.66 | 92.48±3.19 |
| CBAM | 91.58±1.49 | 95.83±0.67 | **99.11±0.18** | 83.91±2.11 | 99.23±0.83 | 92.34±4.15 |
| GCT | 90.07±2.34 | 96.07±0.75 | 98.99±0.18 | 84.74±3.75 | **99.55±0.28** | 93.09±3.24 |
| SRM | 93.04±2.14 | 96.01±0.81 | 98.92±0.17 | 84.15±3.45 | 99.51±0.49 | 91.47±3.62 |
| SNSM ($\gamma$) | 92.89±2.00 | 96.13±0.72 | 98.91±0.16 | **85.19±3.48** | 99.50±0.39 | 92.84±3.64 |
| SNSM ($\gamma$+$\beta$) | **93.33±1.68** | **96.43±0.66** | 98.93±0.16 | 83.60±2.96 | 99.45±0.73 | 92.88±3.12 |
| Range of block means | 3.84 | 0.60 | 0.31 | 1.59 | 0.32 | 1.65 |

the advantage disappears (Table 3). In the random 50/15 split the eight block means span only 0.14 AA points, no difference is significant and every bootstrap interval lies within ±0.3 points. In the standard split they span 0.35 points; the only significant difference, against ECA, is 0.33 points. Because the random 50/15 split keeps the random geometry and changes only the class counts, the 3.84-point spread of the random 5% protocol is explained by the one or two training pixels of the smallest classes, not by spatial leakage.

**Only the smallest classes, and only on Indian Pines.** Changing only the three smallest IP classes confirms this (Table 5): with 15 training pixels for them and 5% for all others, the block means span 0.60 points instead of 3.84. A small edge remains: SNSM ($\gamma$+$\beta$) is still significantly better than every other block, by 0.30–0.60 points, which reaches our 0.5-point threshold only against SE and CBAM (Table 6). The converse test fails, however. Reducing the three smallest classes of PU and KSC to 1, 1 and 2 training pixels lowers AA by about 15 and 7 points, but the dispersion-based gates do not gain: on PU, SNSM ($\gamma$+$\beta$) has the lowest mean AA and is significantly worse than ECA and SNSM ($\gamma$), and on KSC no difference is significant and SRM is the lowest. The Indian Pines effect therefore depends on which classes are scarce, not only on how scarce they are.

**Patch-disjoint splits.** Removing patch overlap lowers AA by 32.0 points on PU and 34.3 on KSC, averaged over blocks, and OA to 55–57% and 74–77% (Table 7). Block means now span 1.80 (PU) and 4.44 (KSC) points, but the seed-to-seed standard deviation is 4.4–7.4 points and the median bootstrap interval is 3.5 (PU) and 3.4 (KSC) points wide, so differences of about two points can be neither confirmed nor excluded. On PU no difference is significant; on KSC, SNSM ($\gamma$+$\beta$) is significantly better than the plain CNN, GCT and SRM, but not than SE, ECA, CBAM or SNSM ($\gamma$), and SE has the highest mean AA (Figure 3). The ranking of blocks also changes between protocols: SRM, tied best on random 5% IP, is the lowest on patch-disjoint KSC.

**Which statistic matters.** Removing descriptors from SNSM on random 5% IP (five seeds; Table 8) shows that the variance carries the gain: with the mean alone AA falls by 1.16 points, and adding the gradient to the mean does not recover it; without the zero initialisation AA is lower on all five seeds. This matches the tie between SNSM

**Table 6.** Paired AA differences SNSM ($\gamma$+$\beta$) − block (points) with 95% bootstrap intervals when only the three smallest classes are changed. *: Holm-corrected two-sided Wilcoxon $p < 0.05$.

| SNSM ($\gamma$+$\beta$) − | IP, smallest 15 | PU, smallest 1/1/2 | KSC, smallest 1/1/2 |
|---|---|---|---|
| Plain CNN | +0.39* [+0.1, +0.7] | -0.59 [-1.5, +0.3] | -0.24 [-0.8, +0.4] |
| SE ($\gamma$) | +0.50* [+0.3, +0.7] | -0.74 [-1.5, +0.0] | +0.79 [+0.2, +1.4] |
| ECA ($\gamma$) | +0.33* [+0.1, +0.6] | -1.19* [-1.8, -0.6] | +0.39 [-0.4, +1.3] |
| CBAM | +0.60* [+0.3, +0.9] | -0.30 [-1.0, +0.4] | +0.54 [-0.5, +1.7] |
| GCT | +0.35* [+0.1, +0.6] | -1.14 [-2.1, -0.2] | -0.21 [-0.9, +0.5] |
| SRM | +0.41* [+0.1, +0.7] | -0.55 [-1.1, +0.1] | +1.41 [+0.5, +2.4] |
| SNSM ($\gamma$) | +0.30* [+0.1, +0.5] | -1.59* [-2.5, -0.6] | +0.03 [-1.1, +1.1] |

**Table 7.** PU and KSC: AA under random 5% splits (five seeds) and OA and AA under patch-disjoint splits (15 seeds) (%, mean ± std). The last row is the range of the block means.

| Block | PU | | | KSC | | |
|---|---|---|---|---|---|---|
| | AA, random | OA, disjoint | AA, disjoint | AA, random | OA, disjoint | AA, disjoint |
| Plain CNN | 99.01±0.16 | 55.31±9.82 | 67.03±5.03 | 99.51±0.36 | 75.13±4.97 | 63.55±6.41 |
| SE ($\gamma$) | 98.80±0.56 | 55.91±9.73 | **67.81±4.39** | 99.43±0.38 | 76.79±4.44 | **67.20±6.66** |
| ECA ($\gamma$) | 99.00±0.20 | 56.30±9.17 | 67.73±4.43 | 99.40±0.66 | 76.40±4.81 | 65.15±5.86 |
| CBAM | **99.11±0.18** | 56.11±10.31 | 66.31±4.80 | 99.23±0.83 | 76.68±4.74 | 66.10±7.39 |
| GCT | 98.99±0.18 | 54.59±9.68 | 66.42±4.57 | **99.55±0.28** | 75.20±5.16 | 63.49±6.80 |
| SRM | 98.92±0.17 | 56.02±9.99 | 66.00±4.73 | 99.51±0.49 | 74.36±4.22 | 62.75±6.38 |
| SNSM ($\gamma$) | 98.91±0.16 | 55.41±9.44 | 67.31±4.57 | 99.50±0.39 | 76.86±4.19 | 65.60±7.15 |
| SNSM ($\gamma$+$\beta$) | 98.93±0.16 | **56.80±11.83** | 67.34±4.95 | 99.45±0.73 | **77.11±3.54** | 67.01±6.00 |
| Range of block means | 0.31 | 2.21 | 1.80 | 0.32 | 2.75 | 4.44 |

and SRM, the only other block that gates on dispersion, and suggests that the spread of a channel's response helps when a class is represented by one or two pixels.

**Cost.** The channel blocks add between 6 (ECA) and 4,256 (SE) parameters to the plain CNN, and SNSM adds 260 (Table 9). SNSM's descriptor computation consists of many small operations, so its batch-1 latency is about 3.7 times and its training time about 1.7 times that of the plain CNN.

## 5. Discussion and limitations

Across four scenes and six protocols, the choice of channel-attention block produced large and consistent differences in only one setting: random 5% splits of Indian Pines. There, the two gates that see channel dispersion were clearly better than SE, ECA, CBAM and GCT, and the controls trace this gap to three classes with one or two training pixels. The same scarcity imposed on Pavia University and KSC did not favour dispersion-based gates, so the effect is specific to those Indian Pines classes rather than a general property of extreme scarcity. Spatial leakage changes the level of accuracy rather than the ranking: once no test patch overlapped a training patch, every block lost more than 30 AA points, and most block differences were no longer significant. Gains of a few AA points reported for a new block on one scene should therefore be checked on other scenes and against the number of training pixels in the smallest classes.

Several limitations apply. The patch-disjoint construction trains on one compact cluster per class, so it adds a spatial shift between training and test data as well as removing overlap, and it leaves few test pixels for some KSC classes; the standard IP split still lets patches overlap, and the standard PU and Houston splits were unavailable. The scarcity stress test reduces the three smallest classes of PU and KSC, which may be easier to separate than the scarce IP classes. PCA is fitted on all pixels

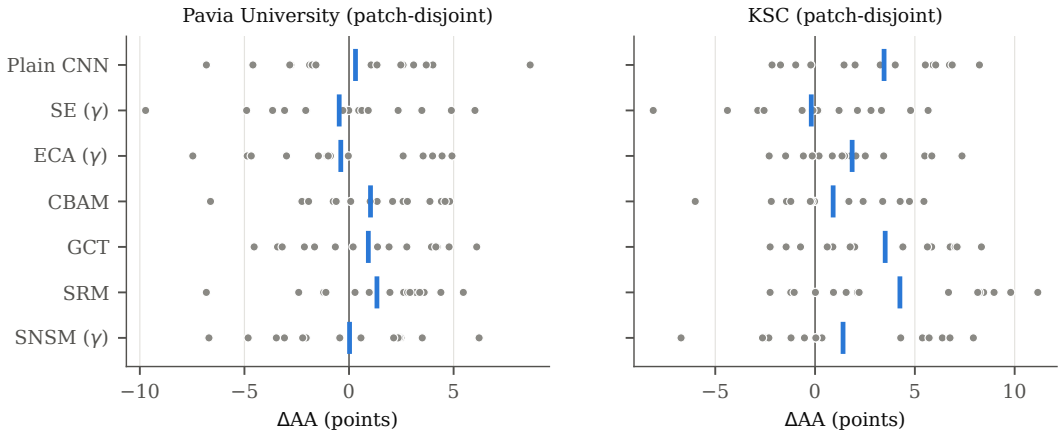



**Figure 3.** Paired same-seed AA differences SNSM ($\gamma$+$\beta$) − block on the patch-disjoint splits. Grey dots are the 15 seeds; the blue bar is their mean.

**Table 8.** SNSM ($\gamma$+$\beta$) ablation on IP, random 5% splits (AA, %; five seeds). ΔAA is the paired difference from the full block; wins count seeds on which the variant is better.

| Descriptors | AA | ΔAA (wins/5) |
|---|---|---|
| mean, var, grad (SNSM) | 94.16±1.07 | – |
| mean | 93.01±0.72 | -1.16 (1/5) |
| mean, var | 94.55±1.67 | +0.39 (4/5) |
| mean, grad | 93.37±1.09 | -0.80 (1/5) |
| mean, var, grad; default init. | 92.67±1.15 | -1.49 (0/5) |

of each scene without labels. All blocks share one shallow backbone, one patch size, one feature pipeline and a fixed schedule; none was tuned, and deeper networks or transformers may use attention differently. HybridSN and SSFTT are less accurate here than in their original papers. With 15 seeds, the patch-disjoint comparisons can only exclude differences larger than about two points.

## 6. Conclusion

For lightweight few-label hyperspectral CNNs, none of seven channel-attention blocks gave a gain that transferred across scenes or protocols. Dispersion-based gates, including the proposed SNSM, added 2–4 AA points on random Indian Pines splits, but only through three classes with one or two training pixels; the advantage shrank to 0.6 points when those classes had 15 pixels, did not appear when the same scarcity was imposed on Pavia University or KSC, and was mostly not significant under patch-disjoint splits, where accuracy dropped by 32–34 points. Class counts, split design and data handling, such as the KSC encoding corrected here, affect reported accuracy far more than the attention block.

## Disclosure statement

The authors report there are no competing interests to declare.

**Table 9.** Model size and cost for a $15 \times 15 \times 60$ IP patch. Training time is the mean per run (100 epochs, random 5% splits).

| Method | Params (M) | MMACs[a] | Latency (ms)[b] | Train IP (s) | Train PU (s) |
|---|---|---|---|---|---|
| Plain CNN | 0.184 | 41.2 | 0.69 | 4.2 | 17.6 |
| SE ($\gamma$) | 0.188 | 41.2 | 1.17 | 4.5 | 19.3 |
| ECA ($\gamma$) | 0.184 | 41.2 | 0.98 | 4.8 | 19.8 |
| CBAM | 0.185 | 41.3 | 1.55 | 5.9 | 24.9 |
| GCT | 0.184 | 41.2 | 1.11 | 4.7 | 20.0 |
| SRM | 0.184 | 41.2 | 1.22 | 4.8 | 20.2 |
| SNSM ($\gamma$) | 0.184 | 41.3 | 2.54 | 7.1 | 29.5 |
| SNSM ($\gamma$+$\beta$) | 0.184 | 41.3 | 2.54 | 7.2 | 30.2 |
| FC ($\gamma$+$\beta$) | 0.190 | 41.2 | 1.31 | 4.7 | 19.8 |
| HybridSN | 1.743 | 137.4 | 0.71 | 26.2 | 115.3 |
| SSFTT | 0.291 | 34.9 | 1.25 | 7.6 | – |

[a] Multiply–accumulates of convolution and linear layers counted by `thop`; tensor operations such as attention products and descriptor statistics are not included. [b] Batch size 1 on a Tesla T4.


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## Data and code availability

The datasets are publicly available (Grupo de Inteligencia Computacional 2020); the standard IP split is distributed with SpectralFormer (Hong et al. 2022). Code, per-run results and the scripts that generate every table and figure are available at `https://github.com/Devguru-codes/research_paper_code`.

## Use of generative AI

Generative AI (Claude, Anthropic) was used to assist with writing experiment code, checking references and editing the manuscript text. The authors reviewed and verified all code, results and text, and take full responsibility for the content.

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